

A Self-Consistent Spatially-Resolved Cluster Dynamics Model for Irradiated Zirconium

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Abstract

Placeholder.

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1. Introduction

Zirconium and its alloys — Zircaloy-2 and Zircaloy-4 in light-water reactor fuel cladding, Zr-2.5Nb in CANDU pressure tubes, and the low-tin variants M5 and ZIRLO in pressurized-water reactor service — operate under prolonged neutron exposure at displacement rates of 10^{-8} to 10^{-7} dpa s⁻¹ and accumulate total damage of 5–30 dpa over a fuel-cycle lifetime [1, 2]. The microstructural changes that result — nucleation and growth of $\langle a \rangle$ prismatic loops, the comparatively delayed appearance of $\langle c \rangle$ -component basal vacancy loops, dislocation channeling under load, anisotropic irradiation growth and creep, and stress-driven hydride reorientation — are the principal source of property degradation in Zr-based components, manifested as hardening, loss of uniform elongation, axial growth of fuel assemblies, and creep-down of pressure tubes [3, 4, 5, 6]. Predicting those changes across the joint envelope of dose, temperature, texture and applied stress is the central quantitative challenge of fuel-element and pressure-tube qualification.

Cluster dynamics (CD), in the form it took in the late 1970s and early 1980s [7, 8, 9, 10, 11], remains the workhorse mesoscopic theory for that prediction. It treats each defect species and size as a population whose concentration evolves under coupled master rate equations balancing in-cascade production, diffusion-controlled reactions, thermal emission and absorption at fixed sinks [12, 13, 14]. Its advantages are direct access to time scales far beyond molecular dynamics [15, 16], explicit resolution of the cluster-size distribution and therefore of every quantity that enters a macroscopic property model [17, 18], and modest cost relative to kinetic Monte Carlo, which together make CD the meeting point between atomistic parameterization [19, 20] and engineering-scale prediction [21, 5].

1.1. The mean-field assumption, and where it fails

Classical CD is a *mean-field, spatially averaged* theory. Every concentration is a single number for the whole material, every sink is represented by a sink strength k^2 that has already been averaged over an assumed random and statistically uniform distribution of absorbers, and the transport that connects a mobile defect to a sink appears only through that average. Three distinct assumptions are folded into this, and each fails under conditions that are not exotic.

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Spatial uniformity.. A sink strength is meaningful only if the diffusion field around each absorber has relaxed to a common far-field value before it meets the next absorber. Near a free surface, a grain boundary, or a denuded band, it has not: the mobile concentration varies by orders of magnitude over a distance comparable with the boundary-layer width, and a single average concentration is not merely imprecise, it does not describe any point in the material. The consequences are structural rather than quantitative. In the calculations reported below, the outermost shell of a 500 nm grain carries essentially the entire prismatic loop population, so a domain average of the loop density is a statement about the boundary layer and not about the material.

Diluteness.. The Avrami-type coalescence kernels that CD uses to represent loop–loop and loop–network interaction are derived by assuming that absorbers are placed independently and at random. Once the volume fraction swept by growing loops becomes appreciable, the placement is no longer independent — loops that have already coalesced are correlated by construction — and the mean-field estimate of the overlap probability ceases to be an estimate of anything. This is the point at which the population must be represented as discrete objects rather than as a density, and locating it is a prediction the model can make about itself.

Isotropy inherited from cubic metals.. Self-interstitial migration in α -Zr is strongly anisotropic, prism-plane mobility exceeding basal mobility by roughly an order of magnitude, and vacancy migration is anisotropic more weakly [19, 20]. Most CD treatments of zirconium nevertheless retain isotropic diffusion constants taken from the body-centered cubic literature and recover the missing directionality through independently fitted bias factors. That conflation is not a book-keeping convenience: it allows a model to believe simultaneously that diffusion is isotropic and that its sinks are biased by the anisotropy of that same diffusion, and it severs the relation between atomic-scale parameters and predicted dimensional change.

1.2. Spatially resolved cluster dynamics

Spatially resolved cluster dynamics (SRCD) replaces the averaged concentrations by fields $c(\mathbf{x}, t)$ obeying reaction–diffusion equations on a meshed domain, so that the transport a mean-field sink strength summarizes is instead solved for [22, 23, 24]. Three things follow that a mean-field model cannot supply.

First, the boundary layer is resolved rather than parameterized. A grain boundary in mean-field CD is a number, $k_{gb}^2 \sim 6\sqrt{k^2/d}$; in SRCD it is a Dirichlet condition, and the depleted zone in front of it — whose width is set by the competition between the local diffusivity and the local sink density, and which therefore differs for each mobile species and evolves through the transient — is an output. The denuded zones seen in irradiated microstructures are a direct consequence, and they are among the few observations that speak to the transport parameters rather than to the reaction parameters.

Second, quantities that are degenerate in a mean-field model become distinguishable. Section 3.5 shows that diffusional anisotropy difference opens a window of the mobile flux ratio over which prismatic interstitial and basal vacancy loops grow together, but that the window is a condition on *one number*; a mean-field model carries that number once, while a spatially resolved model carries it as a field, and the observed coexistence of both loop characters can then be a statement about different points of the same grain rather than about a single balance holding everywhere.

Third, the fields are the natural input to a discrete model. Dislocation dynamics, micromechanical models of channeling, and fracture models all require defect positions and sizes, not densities. A spatially resolved continuum field carries the information a discrete population must reproduce — local number, local size, local orientation — and the conversion between the two becomes a well-posed problem with conservation laws attached, rather than a rescaling of a single average [25, 26, 27].

1.3. Self-consistency

The model presented here is self-consistent in a specific and restrictive sense. Every capture efficiency that appears in a loop growth law, and every sink strength that the same loops impose on the mobile species, is generated from the *same* anisotropic diffusion tensors that transport those species, through the tensorial form of the diffusional-anisotropy-difference (DAD) correction [28, 29, 30, 31]. There is no independent set

of phenomenological bias factors. One consequence is immediate and is the reason the constraint is worth imposing: a change of the diffusion anisotropy now moves the growth laws and the sink strengths together, in the directions the transport dictates, and cannot be absorbed by refitting a bias.

The immobile population is correspondingly expanded, from the lumped stress-aligned pair of the earlier mean-field model to nine crystallographically resolved families: the three prismatic variants a_1, a_2, a_3 of both interstitial and vacancy character, and a three-state basal vacancy chain running from the stacking-fault pyramid through the faulted loop to the perfect loop. The expansion follows the experimental loop character [32, 33, 34, 35, 36]: prismatic $\langle a \rangle$ loops of both signs, and basal $\langle c \rangle$ -component loops that are predominantly *faulted* and that form by collapse of a pyramidal precursor rather than by accretion. Each family carries the first three moments of its size distribution, so that the width of the distribution — not only its mean — is transported, and the conversion to a discrete population can draw loop sizes from a distribution instead of assigning every loop at a point the same size.

1.4. Contributions and outline

This paper reports the formulation, its numerical solution, and its first application to a spatially resolved single crystal. Specifically:

- (i) a self-consistent SRCD formulation for α -Zr in which capture efficiencies and sink strengths are generated from the diffusion tensors (Sections 2 and 3);
- (ii) a two-time-scale operator split that makes of order 10^6 stiff ordinary differential equations per substep tractable, with the cost argument that justifies it and the splitting error that pays for it (??);
- (iii) a continuum-to-discrete conversion with stated conservation guarantees, and a grain-boundary treatment that makes the boundary a sink for immobile loops as well as for mobile defects (??);
- (iv) a calibration against 78 experimental measurements of loop density and diameter, presented as a maximum *a posteriori* estimate under stated priors, with an explicit account of which parameters the data determine and which the bounds determine (??); and
- (v) a 500 nm hexagonal single-crystal calculation to 10 dpa reporting mobile and immobile fields, the closed atom balance including the boundary channel, the predicted denuded zone, and the discrete loop population the fields imply (??).

Section 2 fixes the defect classes and the reactions among them. Section 3 gives the formulation, beginning from the general master equation and reducing it to the equations actually integrated. ?? is the solution method. ?? is the continuum-to-discrete map and the grain-boundary models. ?? is the calibration. ?? is the single-crystal study, and ?? draws the conclusions and states what remains open. Four appendices collect the capture-efficiency derivation, the moment closure, the peripheral-emission model that replaces fitted annealing lifetimes, and the state-vector layout.

2. Defect Classes and Reactions

2.1. The general class set

The primary irradiation defects in hcp zirconium are vacancies, self-interstitial atoms and their clusters. Because the lattice is elastically and crystallographically anisotropic, a cluster is not specified by its size alone: its Burgers-vector character and habit plane determine its contribution to irradiation growth, to hardening, and to the sink field seen by every other species. The observed loop population in irradiated Zr and its alloys is conventionally grouped into $\langle a \rangle$ loops and $\langle c \rangle$ -component loops [37, 38, 35], with

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad \mathbf{b}_c \parallel [0001], \quad (1)$$

the $\langle c \rangle$ loops carrying a Burgers-vector component along the hexagonal axis.

Four principal cluster families span the population, together with the two free monomers [39, 3, 5, 40, 41]:

$$I_n^a, \quad V_n^a, \quad S_n, \quad V_n^c, \quad (2)$$

where I_n^a is an interstitial prismatic $\langle a \rangle$ loop of n excess self-interstitials on a $\{10\bar{1}0\}$ habit; V_n^a a vacancy prismatic loop of the same habit and Burgers vector but opposite character; S_n a basal stacking-fault pyramid (SFP) of n aggregated vacancies, bounded by basal $\frac{1}{6}\langle 20\bar{2}3 \rangle$ and prismatic facets, formed directly from collapsed cascade vacancy debris; and V_n^c a basal vacancy $\langle c \rangle$ -component loop, which nucleates above a temperature- and dose-rate-dependent threshold, in part by unfaulting of supercritical SFPs, and which drives breakaway irradiation growth at high fluence [42, 43, 40]. The free monomers are the mono-interstitial C_i and the monovacancy C_v . No helium and no three-dimensional void population is carried: Zr alloys generate a negligible (n, α) helium yield, and multi-vacancy cascade debris in α -Zr collapses onto basal or prismatic habits rather than forming compact cavities [3, 5].

The classification separates *character* — interstitial or vacancy — from *crystallography* — prismatic, basal pyramidal, or basal planar — and both distinctions are load-bearing. Two families that share a habit plane and a Burgers vector but differ in character, I_n^a and V_n^a , respond to the same mobile fluxes with opposite sign; two families that share a character but differ in habit, V_n^a and V_n^c , present differently oriented capture cross-sections to an anisotropic diffusion field and therefore compete for the same vacancies on unequal terms. It is that second competition, taken up in Section 3.5, that the self-consistent formulation exists to describe without a free parameter.

2.2. The reduced set actually integrated

Equation (2) is size-resolved: each family is an infinite ladder in n . The model solved here replaces the ladder by a small number of *moments* of the size distribution per family (Section 3.7) and resolves instead the crystallographic *variants* that a size-resolved but variant-lumped model cannot distinguish. Writing the character as $s \in \{i, v\}$ and the habit as $k \in \{a_1, a_2, a_3, c_0, c_f, c_p\}$, the families carried are

$$\mathcal{F} = \underbrace{\{(i, a_1), (i, a_2), (i, a_3)\}}_{\text{prismatic interstitial}} \cup \underbrace{\{(v, a_1), (v, a_2), (v, a_3)\}}_{\text{prismatic vacancy}} \cup \underbrace{\{(v, c_0), (v, c_f), (v, c_p)\}}_{\text{basal vacancy chain}}, \quad (3)$$

nine in all, of which eight are dislocation loops and one is not: c_0 is the stacking-fault pyramid, the precursor from which the basal loops form. Write $\mathcal{F}_L = \mathcal{F} \setminus \{(v, c_0)\}$ for the eight loop families; every statement below that invokes a perimeter, a line density or a coalescence channel is a statement about $\mathcal{F}_L$.

Three features distinguish Eq. (3) from the lumped, stress-aligned pair of the earlier mean-field model, and each is a statement about crystallography rather than about the solver.

1. **The prismatic loops are resolved into their three variants a_1, a_2, a_3 , of both characters.** The variants carry their own habit normals $\hat{\mathbf{a}}_k$, so the resolved stress on each is computed rather than parameterized, and the response to an arbitrary stress state is a redistribution among the three rather than a shift of one scalar. At zero deviatoric stress they are equivalent by symmetry, receive equal shares, and summing them recovers the lumped population exactly — which is the regression test on the change.
2. **There is no basal interstitial family.** Basal interstitial $\langle c \rangle$ loops are not a reported population in neutron-irradiated Zr. A formal c_i population is retained in the growth inequalities of Section 3.5 for symmetry but is not a state variable.
3. **The basal vacancy population is a three-state chain, $c_0 \rightarrow c_f \rightarrow c_p$.** The pyramid c_0 is not planar and stores its vacancies with a nearly isotropic relaxation volume, whereas the faulted state c_f and the perfect state c_p are planar and store different vacancy content per unit area. Both transitions are thermally activated and both are one-way.

The two habits are mutually *orthogonal*: the normals $\hat{\mathbf{a}}_k$ of the three prismatic variants lie in (0001) , while the basal normal is $[0001]$ itself. That orthogonality is why no single electron-beam direction images

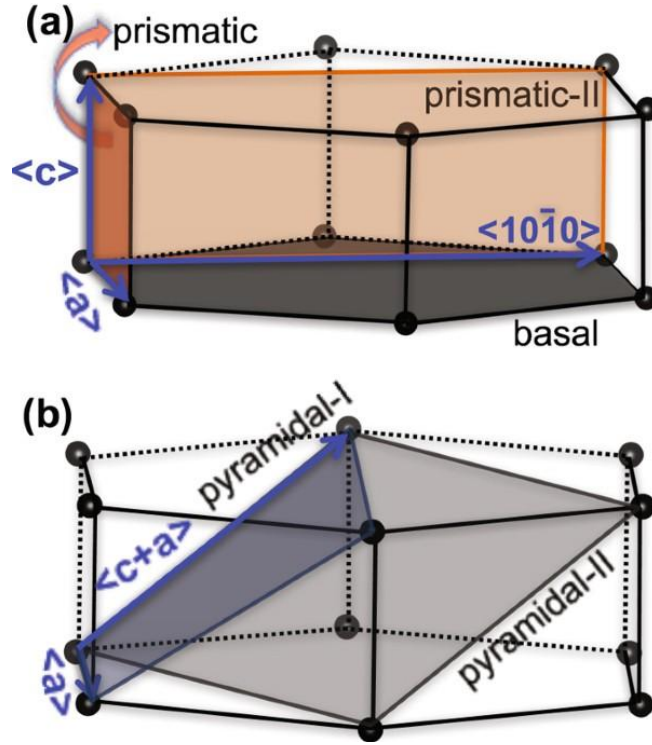


Figure 1: The principal planes and lattice translations of the hcp structure, on which the nine families of Eq. (3) are placed. (a) The *basal* plane (0001) (shaded, bottom) is the habit of c_0 , c_f and c_p ; the first-order *prismatic* plane $\{10\bar{1}0\}$ (the narrow shaded face at the left) is the habit of a_1, a_2, a_3 ; the second-order plane $\{11\bar{2}0\}$ (the wide shaded region) is a diagonal plane through the cell on which $\langle a \rangle$ loops are also observed. The blue arrows are the two shortest lattice translations, $\langle a \rangle = \frac{1}{3}\langle 11\bar{2}0 \rangle$ in the basal plane and $\langle c \rangle = [0001]$ along the cell axis. (b) The inclined pyramidal-I $\{10\bar{1}1\}$ and pyramidal-II $\{11\bar{2}2\}$ planes contain $\langle c+a \rangle = \frac{1}{3}\langle 11\bar{2}3 \rangle$ and carry *no* family of Eq. (3): $\langle c+a \rangle$ is the glide vector that accommodates c -axis strain in deformation, and the planes containing it are not observed habit planes for irradiation-induced loops. The absence of a pyramidal family is therefore an experimental statement and not a truncation. The cell is drawn at an arbitrary axial ratio; α -Zr has $c/a \simeq 1.5944$.

both populations at true size — a zone axis showing one family face-on views the other edge-on — so any comparison of Eq. (3) against TEM densities must combine at least two zone axes, and a single-axis micrograph constrains at most one habit. It is also why the calibration of ?? treats the $\langle a \rangle$ and $\langle c \rangle$ data as separate target sets rather than as one.

2.3. Geometry: from content to radius

Each family carries a character sign η_s (+1 interstitial, -1 vacancy) and three spatially resolved fields: the *cluster density* $n_{sL}^k(\mathbf{x})$, clusters per host atom; the *defect content* $c_{sL}^k(\mathbf{x})$, point defects stored in that family per host atom; and the *second content* $q_{sL}^k(\mathbf{x})$ of Section 3.7, which carries the width of the size distribution. All three are dimensionless, so a number density per unit volume is n/Ω and a stored-defect density is c/Ω , with Ω the atomic volume.

The quantity that converts loop area into stored defects is not $|\mathbf{b}|$ but the edge component $\mathbf{b} \cdot \mathbf{n}$: a loop of radius R holds $\pi R^2(\mathbf{b} \cdot \mathbf{n})/\Omega$ defects. The mean radius, the mean size and the line density of family (s, k) therefore follow from its content and density as

$$R_{sL}^k = \lambda_k \sqrt{\frac{c_{sL}^k}{n_{sL}^k}}, \quad \lambda_k = \sqrt{\frac{\Omega}{\pi(\mathbf{b} \cdot \mathbf{n})_k}}, \quad m_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k, \quad (4)$$

with one geometric length per habit: λ_a shared by all six prismatic families, and λ_{cf} , λ_{cp} for the two basal states. Equation (4) describes a planar object and does not apply to the pyramid, whose stored vacancies fill a volume rather than bound an area; its size follows from the volumetric relation

$$R_{vL}^{c_0} = \left(\frac{m_{vL}^{c_0} \Omega}{\sqrt{8}} \right)^{1/3}, \quad (5)$$

so the pyramid carries no λ_k , no Burgers vector and no line density. What it does carry is a capture cross-section, and Section 3 shows that a cross-section is the only thing either the growth law or the sink strengths ever ask of a family. Table 1 lists the nine families with the crystallography that fixes their geometry.

Table 1: The nine families of Eq. (3), with the crystallography that fixes each one's geometry. $\lambda_k = \sqrt{\Omega/\pi(\mathbf{b} \cdot \mathbf{n})_k}$ is the length converting $\sqrt{m}$ into a radius, Eq. (4). The lattice constants are $a = 0.323$ nm, $c = 0.515$ nm, hence $c/a = 1.5944$ — the physical ratio and not the ideal $\sqrt{8/3} = 1.6330$ — and $\Omega = (\sqrt{3}/4)a^2c = 2.3266 \times 10^{-29}$ m³.

Family	Habit	$\mathbf{b}$	Character	λ_k (nm)	Comment
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3}\langle 11\bar{2}0 \rangle$	i	0.151	The dominant early population; $\mathbf{b} \cdot \mathbf{n} = a$.
a_1, a_2, a_3	$\{10\bar{1}0\}$	$\frac{1}{3}\langle 11\bar{2}0 \rangle$	v	0.151	Same habit and $ \mathbf{b} $ as the interstitial variants; separated from them only by capture character.
c_f	(0001)	$\frac{1}{6}\langle 20\bar{2}3 \rangle$	v	0.170	Faulted, $\mathbf{b} \cdot \mathbf{n} = c/2$, one removed basal layer. The dominant observed $\langle c \rangle$ loop.
c_p	(0001)	[0001]	v	0.120	Perfect, $\mathbf{b} \cdot \mathbf{n} = c$, two removed layers, hence $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$.
c_0	(0001) base	partials	v	—	The stacking-fault pyramid: not planar, so Eq. (4) is replaced by Eq. (5) and it carries no λ_k . Precursor to c_f .

2.4. Reaction classes

The coupled kinetics are generated from a directed multigraph — the *reaction admissibility graph* (RAG) — whose vertices are the admissible cluster states and whose edges are the elementary reactions permitted by the polarity, habit-plane and mobility constraints above. The master equations follow by a deterministic

graph walk: at each vertex the time derivative is the signed sum of the rate kernels of all incoming (gain) and outgoing (loss) edges. Vertices partition by polarity,

$$\mathcal{V}^+ = \{C_i\} \cup \{C_{2i}, C_{3i}\} \cup \{(i, a_k)\}, \quad (6)$$

$$\mathcal{V}^- = \{C_v\} \cup \{(v, a_k)\} \cup \{(v, c_0), (v, c_f), (v, c_p)\}, \quad (7)$$

so the interstitial prismatic families lie on the positive branch and the six vacancy families on the negative one. Table 2 catalogs the edge classes; Fig. 2 draws the graph for the family set of Eq. (3).

Table 2: Admissible process classes of the irradiated-Zr reaction admissibility graph. Each elementary reaction maps to one abstract edge class and one rate kernel; the explicit kernels appear in Section 3 at the equation indicated. P10 has no counterpart in the mean-field treatment and is introduced in ??.

Process	Abstract edge class	Kernel	Given in
P1: vacancy–SIA recombination	cross-polarity (annihilation)	$R_{i,v}$	Eq. (11)
P2: monomer capture by loop or SFP	growth / shrinkage	ϕ_{km}	Eq. (22)
P3: cluster–cluster coalescence	same-polarity	$\varphi_{LL}, \varphi_{LN}$	Eq. (25)
P4: absorption at the network	sink edge	$R_{m,s}$	Eq. (13)
P5: thermal monomer emission	dissociation	$\mathcal{E}_{2i}, \mathcal{E}_{3i}$	??
P6: cascade defect production	source edge	G_m, G_k	??
P7: SFP $\rightarrow c_f$ collapse, $c_f \rightarrow c_p$ unfaulting	inter-population transfer	ν_{col}, ν_{uf}	Eq. (27)
P8: loop absorbed by the network	network-evolution edge	φ_{LN}	Eq. (25)
P9: absorption at a grain boundary, mobile	Dirichlet boundary	—	Eq. (15)
P10: absorption at a grain boundary, <i>loops</i>	size-gated removal	ν_{gb}	??

3. Cluster Dynamics Formulation

3.1. From the master equation to the reduced system

The general cluster-dynamics master equation for a population indexed by a species label α and a size n is a balance of gains and losses over the edges of the graph of Fig. 2,

$$\frac{\partial C_n^\alpha}{\partial t} = -\nabla \cdot \mathbf{J}_n^\alpha + G_n^\alpha + \sum_{\beta, n'} [\mathcal{K}_{n-n', n'}^{\alpha\beta} C_{n-n'}^\beta C_{n'}^{\beta'} - \mathcal{K}_{n, n'}^{\alpha\beta} C_n^\alpha C_{n'}^\beta] + \varepsilon_{n+1}^\alpha C_{n+1}^\alpha - \varepsilon_n^\alpha C_n^\alpha - \mathcal{D}_n^\alpha C_n^\alpha, \quad (8)$$

with $\mathbf{J}_n^\alpha = -\mathbf{D}_n^\alpha \nabla C_n^\alpha$ the diffusive flux, G_n^α the cascade source, $\mathcal{K}$ the pairwise reaction kernels, ε the thermal emission rates, and $\mathcal{D}$ the absorption rate at fixed sinks. Equation (8) is one equation per (α, n) and per mesh node, and is not solvable as written for the size ranges of interest.

Two reductions bring it to the system integrated here, and they cut in different directions.

Mobile species: truncate the ladder.. Only the monomers and the smallest interstitial clusters migrate. The mobile set is closed at $\mathbf{m}_M = [-1, +1, +2, +3]^T$ — monovacancy, mono-, di- and tri-interstitial — with the signed entries m^k carrying the polarity $p^k = m^k/|m^k|$. Everything larger is immobile and is carried by the second reduction. The mobile block retains its spatial derivative, because transport is what it is for.

Immobile families: truncate the moments, keep the variants.. For the immobile families the ladder in n is replaced by the first three moments of the size distribution of each family in Eq. (3),

$$n_{sL}^k = \int f_k dm, \quad c_{sL}^k = \int m f_k dm, \quad q_{sL}^k = \int m^2 f_k dm, \quad (9)$$

closed by the log-normal ansatz of Section 3.7. The immobile block loses its spatial derivative entirely: a loop does not diffuse, and the only thing that couples one node’s loops to another’s is the mobile field. That is not an approximation — it is the property that makes the numerical method of ?? possible.

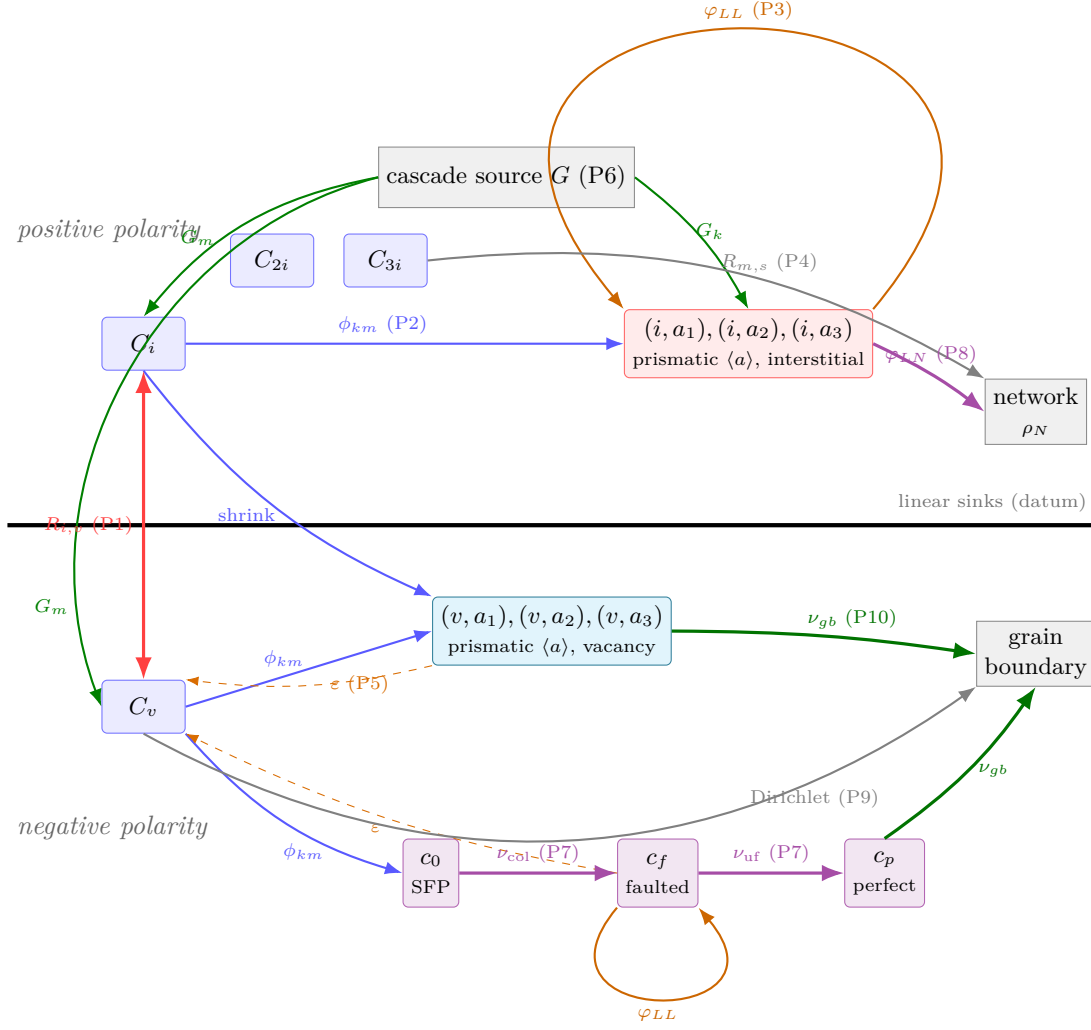


Figure 2: Reaction admissibility graph for the nine families of Eq. (3). Vertices above the datum carry positive (interstitial) polarity, those below it negative (vacancy) polarity. Blue arrows are monomer capture (P2), which acts on a vacancy loop with the opposite sign when the captured species is an interstitial; dashed orange is thermal emission (P5); the heavy red double arrow is recombination (P1); orange loops are coalescence (P3); violet is the one-way basal chain $c_0 \rightarrow c_f \rightarrow c_p$ (P7) and the loop–network transfer (P8); gray is absorption at the network (P4) and the Dirichlet condition on the mobile species at a grain boundary (P9). The green edges (P10) are the channel introduced in ??, by which a loop that has grown large enough to intersect a boundary is absorbed by it — an edge with no counterpart in a mean-field treatment, where a loop has no size relative to a boundary that is itself only a sink strength. The three prismatic variants of each character are drawn as one vertex for legibility; they are separate populations that exchange nothing with one another and differ only in habit normal.

3.2. The mobile block

Because the mobile species reach quasi-steady state on a time scale $\tau_M \sim (k^2 D)^{-1}$ of order microseconds, while the immobile families evolve on the dose scale (1 dpa = 10^7 s at $G = 10^{-7}$ dpa s $^{-1}$), the mobile block is solved as a *steady* reaction–diffusion problem for the immobile state currently held. Writing $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$ in per-atom units,

$$\mathbf{0} = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}, \quad (10)$$

where upright sans-serif bold denotes arrays in *species space* and bold italic ordinary Cartesian vectors and tensors, so that the entries $\mathbf{J}^x = -\mathbf{D}^x \nabla c^x$ of the stacked flux $\mathbf{J}$ are themselves Cartesian vectors and $\nabla \cdot$ acts on each.

The terms are, in order: the cascade production $\mathbf{P}_M = G\varepsilon \mathbf{f}$, with G the displacement rate, ε the surviving efficiency and $\mathbf{f}$ the partition of production among the mobile species; the second-order reactions $\mathbf{R}_2$, whose entries are the pairwise rate constants

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega} (\bar{D}^m + \bar{D}^n)(r^m + r^n), \quad r^m = \begin{cases} \left(\frac{3\Omega}{4\pi} \right)^{1/3}, & |m^m| = 1, \\ \left(\frac{|m^m| \Omega}{\pi b} \right)^{1/2}, & |m^m| > 1, \end{cases} \quad (11)$$

with the recombination-volume factor β_r carried on the interstitial–vacancy channel and the combinatorial factors retained on the like-species channels; the thermal emission matrix $\mathbf{E}$, whose entries are $\alpha^{m,n} = \beta^{m,n} \exp(-E_{m,n}^b/k_B T)$; the linear absorption matrix $\mathbf{D}$; and $\mathbf{S}_{vL}$, the vacancies returned to the pool by peripheral emission from vacancy loops (??).

The absorption matrix is where the immobile population acts back on the mobile one, and it is written so that no defect can be counted twice:

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_L, \quad \mathbf{D}_d = \text{diag}\{\bar{D}^v, \bar{D}^i, \bar{D}^{2i}, \bar{D}^{3i}\}, \quad (12)$$

whose first term carries the *network alone*,

$$C_s^v = \frac{a^2}{z_c} \rho_N, \quad C_s^i = \frac{a^2}{z_c} Z_N \rho_N, \quad (13)$$

and whose second carries the loops through the diagonal sink-strength block

$$\mathbf{K}_L = \text{diag}\{\mathbf{k}^2\}, \quad k_m^2 = \sum_{(s,k) \in \mathcal{F}_L} Z_{sk,m} \rho_{sL}^k + Z_{vc_0,m} \alpha_{\text{sfp}} \pi R_{vL}^{c_0} \frac{n_{vL}^{c_0}}{\Omega} + Z_{N,m} \rho_N. \quad (14)$$

The loops must not also appear in Eq. (13), or their absorption would be counted once as a network sink and once as loop growth. The middle term of Eq. (14) is the stacking-fault pyramid, which has no line density and enters through a compact capture cross-section instead; Eq. (24) below shows that a cross-section is the only thing either side ever asks of a family.

Boundary conditions. On a free surface or a grain boundary the mobile species are pinned at thermal equilibrium,

$$c^v = c^{v,\text{eq}} = \exp(-E_v^f/k_B T), \quad c^i = c^{2i} = c^{3i} = 0 \quad \text{on } \partial\Omega_D, \quad (15)$$

which is process class P9 of Table 2. This replaces the mean-field grain-boundary sink strength $k_{gb}^2 \simeq 6 \sqrt{k^2}/d$ by a boundary condition, and the depleted layer in front of it becomes an output rather than a parameter. Faces declared periodic instead carry periodic conditions on all four species.

3.3. Anisotropic transport and the single anisotropy parameter

Each mobile species carries a diffusion tensor $D_{ij}^x = D_{0ij}^x \exp(-E_{ij}^x/k_B T)$, diagonal in the crystal frame with the basal components equal and the c -axis component distinct. The whole of that anisotropy is expressed by one number per species,

$$p^m = \left(\frac{D_{\parallel}^m}{D_{\text{basal}}^m} \right)^{1/6}, \quad (16)$$

and the migration energies are generated from it at *fixed* effective diffusivity,

$$E_{\langle 11 \rangle}^m = E_{\langle 22 \rangle}^m = E_{\text{eff}}^m + 2k_B T \ln p^m, \quad E_{\langle 33 \rangle}^m = E_{\text{eff}}^m - 4k_B T \ln p^m, \quad (17)$$

so that the orientation-averaged diffusivity $\bar{D}^m = (\det \mathbf{D}^m)^{1/3}$ is exactly the scalar D_m of the mean-field kinetics. **No mobility is redefined by the anisotropy — only its directional weighting.** This is the technical device that makes the formulation self-consistent: Eqs. (16) and (17) are a change of variable in which p^m is the only anisotropy the model has, and Section 3.4 uses that same p^m and no other parameter.

3.4. Capture efficiencies

Every capture efficiency is a product of factors, one per degeneracy to be broken:

$$Z_{sk,m}(\sigma) = \underbrace{Z_m^0}_{\text{elastic bias}} \underbrace{A_{h(k)}(p^m)}_{\text{DAD, orientation}} \underbrace{X_{s,m}}_{\text{character}} \underbrace{\left[1 + \frac{\Delta Z_{h(k),m}^{\text{SIPA}}(\sigma_{kk})}{Z_m^0 A_{h(k)}(p^m)} \right]}_{\text{stress, SIPA}}, \quad (18)$$

with $h(k) \in \{a, c\}$ the habit of family k and the Woo diffusional-anisotropy factors [28, 29, 31]

$$A_a(p) = \frac{p + p^{-2}}{2}, \quad A_c(p) = p. \quad (19)$$

Each factor breaks exactly one degeneracy and is silent on the others. $A_{h(k)}$ depends on the habit alone — on the angle between the loop normal and the c axis — and therefore assigns one value to all six prismatic families and one to all three basal states, the pyramid included, whose base is (0001) so that $h(c_0) = c$. It separates prismatic from basal but is degenerate in character. $X_{s,m}$ is the character factor, which separates an interstitial from a vacancy loop of the same habit [44, 45]; the coexistence window it opens has log-width $\ln \chi$ with $\chi = X_{iI} X_{vV} / (X_{iV} X_{vI})$.

Equation (18) contains no free bias. $A_{h(k)}$ is built from the same p^m that Eq. (17) puts into the diffusion tensors, so a change in the transport anisotropy moves the growth laws and the sink strengths of Eq. (14) together. The two remaining constants, Z_m^0 and $X_{s,m}$, are calibrated (??); the orientation dependence is not.

The forms in Eq. (19) are not monotone in the same way, and the consequence is quantitative and easily missed. $A_c(p) = p$ increases without bound, while $A_a(p) = (p + p^{-2})/2$ has its minimum at $p = 2^{1/3} = 1.2599$. A single change of p^v therefore pushes the basal and prismatic families in *opposite* directions, starving $\langle c \rangle$ growth while accelerating $\langle a \rangle$ shrinkage, or the reverse. Any statement about the effect of anisotropy that does not distinguish the two habits is incomplete.

3.5. Coexistence, and what a spatially resolved model adds

Define the mobile flux aggregates and their ratio,

$$\Phi_I = \sum_{n \in I_{\text{mob}}} n \bar{D}^{I_n} c^{I_n}, \quad \Phi_V = \bar{D}^v c^v, \quad x \equiv \frac{\Phi_V}{\Phi_I}. \quad (20)$$

The net driving force on a family of character sign η_s is $F_s = \eta_s (Z_{s,I} \Phi_I - Z_{s,V} \Phi_V)$, so that

$$\text{interstitial family grows} \iff x < \frac{Z_{s,I}}{Z_{s,V}}, \quad \text{vacancy family grows} \iff x > \frac{Z_{s,I}}{Z_{s,V}}. \quad (21)$$

Under identical local fluxes and identical capture coefficients an interstitial and a vacancy loop of the same orientation therefore *cannot* both grow. The observed coexistence of both characters is thus a statement that at least one idealization has failed.

Diffusional anisotropy resolves part of it. Because A_a and A_c differ, the prismatic and basal thresholds in Eq. (21) differ, and there is a finite window of x over which prismatic interstitial and basal vacancy loops grow together; the window has width $g(p^v)/g(p^i)$ with $g(p) = 2/(1 + p^{-3})$, strictly increasing, so it is open exactly when $p^i < p^v$. But DAD is degenerate in *character*: $A_{h(k)}$ knows the habit and nothing else, so it cannot by itself support the observed coexistence of interstitial and vacancy loops on the *same* prismatic variant.

Two resolutions are carried, and the first is what a spatially resolved model supplies for free. Equation (21) is a condition on one number, and in a mean-field model that number exists once for the whole material. In SRCD, x is a *field*: it varies by orders of magnitude between the interior and the boundary layer, because c^v and c^i have different diffusivities and therefore different boundary-layer widths. Coexistence can then be a statement about different points of the same grain rather than about one balance holding everywhere. The second resolution is the character factor $X_{s,m}$ of Eq. (18), which enters as a re-parameterization of the efficiencies and not as a structural change.

3.6. The immobile block

Growth.. All nine families obey one growth law, in which the stored content changes under the net biased flux of the correct sign:

$$\dot{c}_{sL}^k|_{\text{growth}} = \Pi_{sL}^k \eta_s \sum_{m \in \{v,i,2i,3i\}} |m| \bar{D}^m Z_{sk,m} c^m \varsigma_s(m), \quad (22)$$

with $\varsigma_s(m) = +1$ when the polarity of species m matches the character s and -1 otherwise. The geometric prefactor is

$$\Pi_{sL}^k = \frac{\lambda_k}{\ell} \varsigma_k^{\text{sink}} \sqrt{n_{sL}^k c_{sL}^k}, \quad \ell \equiv \frac{z_c \Omega}{2\pi a^2}, \quad (23)$$

and because $\rho_{sL}^k = 2\pi\lambda_k \sqrt{n_{sL}^k c_{sL}^k}$ by Eq. (4),

$$\Pi_{sL}^k = \varsigma_k^{\text{sink}} \frac{\rho_{sL}^k}{2\pi\ell} \quad \text{identically, for every } (s,k) \in \mathcal{F}_L. \quad (24)$$

The prefactor in the growth law is the family's own sink density divided by $2\pi\ell$. That identity is what makes Eq. (22) and Eq. (14) two readings of one quantity, and it is what carries the growth law onto the pyramid unchanged: a family needs no geometry beyond the capture cross-section it presents, and the pyramid supplies that cross-section as $\Pi_{vL}^{c_0} = \varsigma_{c_0}^{\text{sink}} \alpha_{\text{sfp}} R_{vL}^{c_0} n_{vL}^{c_0} / 2\ell$.

Nucleation.. Cascade production deposits directly into the immobile families at a rate $G_k = G \epsilon_k$ per family, each new cluster carrying m_k^{nuc} defects, so that $\dot{n}_{sL}^k|_{\text{nuc}} = J_{sL}^k = G_k / m_k^{\text{nuc}}$ and $\dot{c}_{sL}^k|_{\text{nuc}} = G_k$. The partition ϵ_k and the nucleation sizes are calibrated quantities.

Coalescence.. Loops are removed by overlap with other loops and with the network. With $N = n/\Omega$, $R = \lambda \sqrt{c/n}$, and the radii capped at the relevant spacing ($R_{LL} = \min(R, (\Omega/n)^{1/3})$, $R_{LN} = \min(R, \rho_N^{-1/2})$), the Avrami overlap probabilities and rate scales are

$$\begin{aligned} \varphi_{LL} &= 1 - \exp(-\kappa_{LL} \frac{4}{3} \pi R_{LL}^3 N), & \varphi_{LN} &= 1 - \exp(-\kappa_{LN} \pi R_{LN}^2 \rho_N), \\ \nu_{LL} &= c_{LL} v_{\text{abs}} N^{1/3}, & \nu_{LN} &= c_{LN} v_{\text{abs}} \sqrt{\rho_N}, \end{aligned} \quad (25)$$

giving the number and content losses

$$\Delta N_{LN} = \nu_{LN} \varphi_{LN} n, \quad \Delta N = \nu_{LL} \varphi_{LL} n + \Delta N_{LN}, \quad \Delta C = \nu_{LN} \varphi_{LN} c. \quad (26)$$

The like-loop channel merges two loops into a larger one and therefore *conserves content*, removing density only; the loop–network channel removes both, and the removed content is booked as network sink absorption so the point-defect balance stays closed. The rate scale is set by the *absorbed*-flux climb speed v_{abs} rather than the net growth velocity, so coarsening persists at saturation. The capping is what makes Eq. (25) saturate rather than diverge as the number density collapses, and $\varphi_{LL} \rightarrow 1$ is the model’s own statement that its mean-field treatment of coalescence has expired — the detector used in ??.

Emission.. Vacancy loops exchange vacancies across their own perimeter at a rate set by a size-dependent binding energy computed from the anisotropic loop self-energy (??), one expression serving both characters, so that the same mechanism shrinks a vacancy loop and grows an interstitial one and the classical climb threshold emerges rather than being imposed. This channel replaces the two fitted annealing lifetimes of the mean-field model. It returns vacancies to the mobile pool through $\mathbf{S}_{vL}$ in Eq. (10).

The basal chain.. The three basal states are linked by two one-way activated transfers,

$$c_0 \xrightarrow{\nu_{\text{col}}} c_f \xrightarrow{\nu_{\text{uf}}} c_p, \quad (27)$$

both size-gated, and both conserving vacancies exactly: the content leaving one state enters the next, and only the *number* changes, because the states store different vacancy content per unit area (Table 1) [46, 47, 45]. The chain is basal and has no prismatic counterpart: *ab initio*-parameterized formation energies place the prismatic vacancy $\langle a \rangle$ loop *below* both the compact cavity and the faulted basal loop, so the planar prismatic loop is already the ground-state geometry rather than something a compact cluster must convert into [47]. **None of the basal-chain rates is calibrated**; the formulation gives their Arrhenius form and no measurement in this work supplies the barriers, so the chain is switched off in the calculation of ?? and its consequences are discussed in ??.

3.7. The size distribution and its closure

Carrying only n and c assigns every loop at a point the same size, and the population that produces is monodisperse by construction. The second content q_{sL}^k removes that restriction. Its Fokker–Planck part is

$$\dot{q}_{sL}^k|_{\text{FP}} = 2\Pi_{sL}^k \left[\underbrace{\Sigma_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8}}_{\text{drift}} + \underbrace{\frac{1}{2} \Sigma_k^{\text{gro}} \Delta_k^{-1/8}}_{\text{spreading}} \right], \quad (28)$$

to which nucleation, the floor current and the basal transfers are appended weighted by m^2 rather than m^1 or m^0 :

$$\dot{q}_{sL}^k = \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\text{min}}^2 \Phi_{\text{min}}^k \pm \Gamma_{\theta}^q. \quad (29)$$

The spreading term does not vanish when the family stops growing: at $\Sigma_k^{\text{net}} = 0$ the drift term is zero and $\dot{q}_{sL}^k = \Pi_{sL}^k \Sigma_k^{\text{gro}} \Delta_k^{-1/8} > 0$, so the distribution keeps broadening at fixed mean size and fixed number. There is no value of the two lower moments that expresses that, which is the argument for carrying the third.

Closing Eq. (29) with a log-normal $f_k(m)$ determines its two parameters from the three carried moments,

$$\bar{m}_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \Delta_k \equiv \frac{q_{sL}^k n_{sL}^k}{(c_{sL}^k)^2} = e^{s_k^2} \geq 1, \quad s_k^2 = \ln \Delta_k. \quad (30)$$

$\Delta_k \geq 1$ is Cauchy–Schwarz on a non-negative measure, so $\Delta_k < 1$ **is not a loose tolerance but a defect**: it says the three carried numbers are not the moments of any distribution. Keeping the triple realizable places a real constraint on how each channel debits q , and ?? states the three rules that follow — of which the one most easily got wrong is coalescence, because a coalescing loop does not leave the family, it *merges*, so fewer loops holding the same content are *larger* loops and coalescence *raises* q . Debiting q by the removed number times $\langle m^2 \rangle$, the natural-looking reading, inverts the sign of the largest single term in the equation.

The closure reaches the rest of the model in exactly one place, as $\Delta_k^{-1/8}$ on the sink prefactor, and it reaches the discrete conversion of ?? as the distribution loop sizes are drawn from.

3.8. Summary of the equations

The system integrated is two blocks with different structure, and the difference between them is the whole of the numerical method.

Mobile block — four fields, spatially coupled, no time derivative..

$$\mathbf{0} = \nabla \cdot (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M) + \mathbf{P}_M + \frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}, \quad \mathbf{c}_M|_{\partial\Omega_D} = \mathbf{c}_M^{\text{eq}}, \quad (31)$$

$4 \times N_{\text{node}}$ unknowns, nonlinear and algebraic, coupled across the mesh by the divergence term alone.

Immobile block — 27 fields, pointwise, no spatial coupling.. For each family $(s, k) \in \mathcal{F}$ at each node independently,

$$\dot{n}_{sL}^k = J_{sL}^k - \Delta N - \Phi_{\min}^k \pm \Gamma_{\theta}^n - \nu_{gb} \Theta_0^k, \quad (32)$$

$$\dot{c}_{sL}^k = c_{sL}^k|_{\text{growth}} + G_k - \Delta C \pm \Gamma_{\theta}^c - \nu_{gb} \Theta_1^k, \quad (33)$$

$$\dot{q}_{sL}^k = q_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\min}^2 \Phi_{\min}^k \pm \Gamma_{\theta}^q - \nu_{gb} \Theta_2^k, \quad (34)$$

with Θ_j^k the size-gated grain-boundary removal of ?? and Γ_{θ} the basal-chain transfers of Eq. (27). Together with the evolving network density ρ_N and the atom-balance accumulators of ??, this is a system of ordinary differential equations at each node — $3 \times 9 + 1$ fields plus accumulators — in which no neighbor's state appears. **The mobile species are the only thing that couples one node to another**, and freezing them, as ?? does, removes the spatial coupling by construction rather than by approximation.

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